DANIYAL FARHANGI

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EDUCATION

Bachelor's Degree In Computer Engineering (See Transcript) *Sep. 2019 - Aug. 2023 (Expected)*
University of Tehran (First University Among Universities in Iran and 329th in Global Universities) Iran
GPA: 3.88/4 (18.23/20)
GPA of last two years: 4/4
Supervisor : Dr. Kazim Fouladi

RESEARCH INTERESTS

- Machine Learning
- Natural Language Processing
- Computer Vision
- Artificial Intelligence
- Data Science
- Chatbots

AWARDS & ACHIEVEMENTS

- Awarded the University of Tehran M.Sc. fellowship and granted direct admission to the master's program at the University of Tehran without having to take the graduate entrance exam.
- Acknowledged as an exceptionally talented student, placing in the **top 5%** among undergraduates.
- Awarded **Tuition-Waiver** at University of Tehran.
- Ranked **top 1%** in the national university entrance exam (among 150,000 students).

RESEARCH EXPERIENCES

Undergraduate research assistant at the University of Tehran Sep. 2022 - Mar. 2023
Under Supervision of Dr. Kazim Fouladi *University of Tehran*

- Working on sentiment analysis algorithms such as NB, SVM, LR, CNN, LSTM, CNN-LSTM, and BERT, and evaluating and comparing the performance of these algorithms against each other.

Natural Language Processing Intern Aug. 2022 - Jan. 2023
Under Supervision of Dr. Hossein Sameti and Dr. Soroush Gooran *Sharif University of Technology*

- Working on open domain generation-based chatbots.
- Participating in weekly sessions of reading scientific papers.

TEACHING EXPERIENCES

Signals and Systems Feb. 2023 - Jul. 2023
Dr. Hossein Aghababa *University of Tehran*

- Responsibility: Teaching and Grading Projects and Assignments

Computer-Aided Digital System Design Feb. 2023 - Jul. 2023
Dr. Hossein Aghababa *University of Tehran*

- Language: Verilog

- Responsibility: Teaching and Grading Projects and Assignments

Advanced Programming Feb. 2022 - Jul. 2022
Dr. Zahra Movahedi *University of Tehran*

- Language: Java
- Responsibility: Grading Projects and Assignments

WORK EXPERIENCES

Research Assistant at the Deep Learning Laboratory of the University of Tehran Mar. 2023 - *Present*
Deep Learning Lab *Tehran, Iran*

- Working on text analysis in social networks by collecting posts and comments by crawling in nets, then analyzing their sentiments.

Deep Learning and NLP Intern Feb. 2022 - Jun. 2022
Buyroad AI · Internship *Tehran, Iran*

- Worked on Retrieval-based Chatbots using bi-LSTM neural network. (Source Code)
- Worked on Persian Search-engine using Elasticsearch and BERT. (Source Code)

Backend Developer Jun. 2020 - Jan. 2022
Nahira Company *Tehran, Iran*

- Nahira is a startup company developing websites and applications.
- I worked on lots of projects, coding Python and Django as a back-end developer. (Source Code 1, Source Code 2)

ACADEMIC PROJECTS

Text Summarization Jun. 2023
University of Tehran

- An abstractive Text Summarization fine-tuning on the T5 transformer model on the xsum dataset. (Project link)

Chatbot Mar. 2023
University of Tehran

- Open-domain generation-based chatbot using large language models such as T5, Falcon7B, and Llama7B with the Reddit dataset. (Project link)

Object Detection Feb. 2023
University of Tehran

- An object detection system using YOLOv5 on the Road Signs dataset. (Project link)

Intent Recognition Nov. 2022
University of Tehran

- An intent recognition system using BERT to identify the intent of a user's input. (Project link)

Sentiment Analysis Sep. 2022
University of Tehran

- Comparison and evaluation of sentiment analysis algorithms such as NB, SVM, LR, CNN, LSTM, CNN-LSTM on Amazon reviews. (Project link)

Machine Translation Oct. 2021
University of Tehran

- A machine translation system that translates text from Persian to English and English to Persian using attention and transformers. (Project link)

News Text Classification Jul. 2021
University of Tehran

- Persian news text classification from Hamshahri corpus based on Bidirectional LSTM. (Project link)

Word2Vec May. 2021
University of Tehran

- This project aims to learn high-quality word embeddings using the Word2Vec algorithm with the continuous bag of words (CBOW) and skipgram models. (Project link)

SELECTED COURSE PROJECTS

Finding Lane Lines on the Road Jul. 2023
Foundations of Computer Vision *University of Tehran*

- Detect lanes in test images and video using OpenCV library. (Project link)

Cars Parking AI May. 2022
Artificial Intelligence and Expert Systems *University of Tehran*

- Used a* algorithm to automatically find path to control parked cars with AI. (Project link)

OS Scheduling Simulation Jan. 2022
Operating Systems *University of Tehran*

- simulated CPU scheduling with FCFS, RR and Preemptive/Non-Preemptive Priority/SFJ algorithms.
- Used a graphical interface to work with the program and plotted metrics such as waiting, turnaround and response time of CPU simulation (Project link)

Bank System Jun. 2020
Advanced Programming *University of Tehran*

- This is a Java-based banking system project that includes a graphical user interface (GUI) developed with JavaFX. This project allows users to create bank accounts, deposit and withdraw money, and view their account information. The project is built using object-oriented programming principles. (Project link)

SKILLS

AI Professional Skills. 2 years:

- Python (TensorFlow, PyTorch, Keras, Scikit-Learn, NumPy, Pandas, Matplotlib)
- Machine Learning
- Deep Learning (ANN, CNN, RNN, LSTM, GAN)
- Natural Language Processing (Transformers, LLM)

Back-end Skills. 1.5 year:

- Django, REST APIs, GraphQL, Unit Tests
- SQL, PostgreSQL, MongoDB, Redis, ElasticSearch, Kafka
- Web Scraping (Scrapy, Beautifulsoup)
- Docker, Git, Linux (Ubuntu)

Language Skills:

- Persian (Native)
- English (Fluent)

Other Skills:

- C++, Java, JavaScript, TypeScript, Node.js
- Methodologies (Agile, Scrum, Waterfall)
- Product Management Tools (Jira, Trello, ClickUp)
- LaTeX, Word, PowerPoint, Excel

TRAINED COURSES

Deep Learning
- University of Tehran
Jun. 2023

Convolutional Neural Networks
- Coursera
Nov. 2021

Natural Language Processing Specialization
- Coursera
Oct. 2022

Machine Learning, Andrew Ng
- Coursera
Sep. 2021

Natural Language Processing with Deep Learning Course
- Youtube Stanford CS224N
May. 2022

Python
- Utech academy (University of Tehran)
Jan. 2020

VOLUNTEER WORKS

IPRIA 2023 Conference Executive Team Jan. 2023 - Feb. 2023

- The 6th International Conference on Pattern Recognition and Image Analysis (IPRIA 2023) which was Held in the Department of Computer Engineering, University of Tehran (Certification)

MVIP 2020 Conference Executive Team Feb. 2020 - Mar. 2020

- The 11th Iranian and The First International Conference on Machine Vision and Image Processing Held at the Department of Computer Engineering, University of Tehran (Certification)

REFERENCES

- Dr. Kazim fouladi : kfouladi@ut.ac.ir
- Dr. Hossein Aghababa : aghababa@ut.ac.ir
- Eng. Mahdi Seyfipoor : mahdiseyfipoor@ut.ac.ir